25900361

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEECT203 - AI AND MACHINE LEARNING FUNDAMENTALS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions Each question carries 1 marks

1 b) ML algorithms	CO1 U 1
2 a) Simplicity & readability	CO2 K 1
3 b) Predicting house prices from past data	CO3 K 1
4 b) Non-linear transformation	CO4 K 1
5 a) Mimic human intelligence	CO1 U 1
6 a) Artificial Intelligence	CO1 U 1
7 a) Data manipulation	CO2 K 1
8 b) Minimize the sum of squared errors	CO3 U 1
9 a) Weighted sum + Activation	CO4 U 1
10 a) Continuous vs Discrete	CO4 U 1
11 a) Series and Data Frame	CO2 K 1
12 a) High-level API for visuals	CO2 K 1
13 b) Unsupervised learning	CO3 U 1
14 b) Relationship between variables	CO2 U 1
15 c) Forward: prediction, Backward: learning	CO4 U 1
16 b) head()	CO2 A 1
17 a) Removing or transforming	CO2 K 1
18 d) Layered structure	CO4 U 1
19 a) A nerve cell	CO4 K 1
20 d) Python	CO2 K 1

[1x20 = 20]

Part B

Short Answer Questions

Answer any 6 questions Each question carries 5 marks

21) Comparison of KNN vs Decision Tree – training & prediction time difference - **2 MARKS**
(lazy vs eager learner) - **3 MARKS**

22) Definition, need, and process of Feature Engineering - 2 **MARKS**
Selection - **1 MARKS**
extraction - **1 MARKS**
transformation - **1 MARKS**

23) Components of Reinforcement Learning - **1 MARKS**
agent, - **1 MARKS**
environment - **1 MARKS**
policy - **1 MARKS**
reward &value. - **1 MARKS**

24) Definition and importance of Neural Networks - 2 **MARKS**
layers, - **1 MARKS**
weights, - **1 MARKS**
learning ability. - **1 MARKS**

25) Role of Normalization & Standardization in ML - **2 MARKS**
scaling data - **1 MARKS**
z-score - **1 MARKS**
min-max - **1 MARKS**

26) Steps to configure virtual environment in PyCharm - **2 MARKS**
create project → interpreter → install dependencies - **3 MARKS**

27) Example of supervised learning - **2 MARKS**
email spam detection / regression example - **3 MARKS**

28) Difference between high and low learning rates - **2 MARKS**
convergence speed - **1.5 MARKS**
overshooting, stability - **1.5 MARKS**

[6x5 = 30]

Part C

Essay Questions

Answer any 2 questions Each question carries 10 marks

29) Detailed explanation of Gradient Descent; - **4 MARKS**
types – batch - **1.5 MARKS**
stochastic, - **1.5 MARKS**
mini-batch - **1.5 MARKS**
convergence visualization. - **1.5 MARKS**

30) Step-by-step working of K-Means Clustering; - **5 MARKS**
example with dataset; - **2.5 MARKS**
example with centroid update illustration. - **2.5 MARKS**

31) Comparison among AI, ML, DL; - **4 MARKS**
structure, scope, and examples of each; - **3 MARKS**
hierarchy diagram. - **3 MARKS**

32) Importance of Data Visualization; - **4 MARKS**
role of libraries (Matplotlib, Seaborn); - **3 MARKS**
types of plots and insights. - **3 MARKS**

[2x10 = 20]

SCHEME PRPEARED BY

DILEEP AUGUSTINE
DEPT OF ELECTRONICS
JAI BHARATH ARTS & SCIENCE COLLEGE, PERUMBAVOOR
MOB: 9947600897